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Top Skills

Nanotechnology
Research and Development (R&D)
Engineering

Certifications

Critical Thinking for Better Judgment
and Decision-Making

Publications

Experimental investigation of fracture
toughness for treated sisal epoxy
composite

Modification of the adhesion energy
of monolayer graphene on SiO₂

Interface engineering to improve
electrochemical performance of
intermediate-temperature solid oxide
fuel cells

Vertical graphene-decorated carbon
nanofibers establishing robust
conductive networks for fiber-based
stretchable strain sensors

Crystallographic dependence
of adhesion energy of chemical
vapor deposition-grown monolayer
graphene on copper foils

Yonas Tsegaye Megra (Ph.D.)

Postdoctoral Research Fellow in 2D Materials & Molecular
electronics | XJTLU (Work Station) & USTC (Mobile Station) | PhD in
Mechanical Engineering
Suzhou, Jiangsu, China

Summary

I am currently a Postdoctoral Researcher / Research Associate at Xi'an Jiaotong-Liverpool University (XJTLU), where I conduct advanced research on 2D materials, molecular electronics, and nanoelectronic device engineering, with a focus on interfacial phenomena and charge transport. I design and develop novel device architectures, including molecular junctions and graphene-based devices, for applications in sensing, energy, and next-generation electronics. My work involves supervising and mentoring PhD students, guiding research on graphene adhesion, nanocomposites, molecular-scale charge transport, and battery-related technologies, while supporting experimental design, data analysis, and scientific writing. I also prepare and manage research proposals and collaborate with international teams on semiconductor devices, energy storage systems, and nanoscale characterization techniques. Previously, I was a Postdoctoral Fellow at the Gwangju Institute of Science and Technology (GIST), Department of Semiconductor Engineering, focusing on nanomaterial synthesis and characterization. I specialized in creating wafer-scale single- and few-layer graphene, transition metal dichalcogenides (TMDs), and successfully fabricated graphene-based flexible substrates for sensors and devices. Before GIST, I was a Postdoctoral Fellow at Sungkyunkwan University (SKKU), Department of Smart-Fab Technology and Mechanical Engineering, where I conducted cutting-edge research in nanomaterial synthesis and nanoscale characterization, publishing in high-impact journals and presenting at international conferences. I hold a Ph.D. in Mechanical Engineering from SKKU, where I developed novel nanofabrication techniques and studied interfacial phenomena using advanced characterization tools. Additionally, I have experience as a Lecturer at Addis Ababa University, where I taught mechanical engineering courses and collaborated on research projects. I am passionate about education, research, and interdisciplinary innovation, and I aspire to advance

scientific knowledge and technology in the field of nanomaterials and nanoelectronics.

Experience

Xi'an Jiaotong-Liverpool University

Postdoctoral Research Fellow in 2D Materials & Molecular electronics

June 2025 - Present (1 year)

Suzhou

Conduct advanced research on 2D materials, molecular electronics, and nanoelectronic device engineering, with a focus on interfacial phenomena and charge transport. Design and develop novel device architectures, including molecular junctions and graphene-based devices, for applications in sensing, energy, and next-generation electronics.

Lead and contribute to high-impact publications, including both review and experimental studies in leading journals. Supervise and mentor PhD students, guiding research on graphene adhesion, nanocomposites, molecular-scale charge transport, and battery technologies, while supporting experimental design, data analysis, and scientific writing.

Prepare and manage research proposals and grant applications aligned with emerging interdisciplinary areas such as AI-enabled sensors and advanced packaging. Collaborate with international research teams on semiconductor devices, energy storage systems, and nanoscale characterization.

Utilize advanced experimental techniques, including CVD synthesis, surface and interface characterization, and electrical transport measurements.

Addis Ababa University

10 years 9 months

Lecturer

July 2017 - Present (8 years 11 months)

Addis Ababa, Ethiopia

Assistant Lecturer

September 2015 - June 2017 (1 year 10 months)

Addis Ababa, Ethiopia

Gwangju Institute of Science and Technology

Postdoctoral Research Fellow, Department of Semiconductor Engineering

April 2024 - June 2025 (1 year 3 months)

South Korea

- Synthesized and characterized 2D materials for next-gen semiconductor applications.
- Developed AI-integrated sensors for enhanced device performance.
- Mentored junior researchers and collaborated with interdisciplinary teams.

Sungkyunkwan University

6 years 9 months

Postdoctoral Research Fellow, Department of Smart-Fab Technology
November 2022 - April 2024 (1 year 6 months)

Suwon, South Korea

Postdoctoral Research Fellow, School of Mechanical Engineering

August 2022 - November 2022 (4 months)

Suwon, Gyeonggi-do, Korea

Researcher, PhD Candidate

February 2019 - August 2022 (3 years 7 months)

Suwon, Gyeonggi-do, Korea

Research Assistant

August 2017 - August 2022 (5 years 1 month)

Suwon, Gyeonggi-do, Korea

Unity University

Lecturer

March 2016 - March 2017 (1 year 1 month)

Education

Sungkyunkwan University

Doctor of Philosophy - PhD, Mechanical Engineering · (August 2017 - August 2022)

Addis Ababa University

Master's degree, Mechanical Engineering · (2015 - 2017)

Addis Ababa University

Bachelor's degree, Mechanical Engineering · (2010 - 2015)